

Maya Beach

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EDUCATION

Carnegie Mellon University

Bachelor of Science in Mechanical Engineering | Minor in Soft Technologies | December 2026

Overall GPA: 3.86/4.0

SKILLS

Technical Skills: SolidWorks, Onshape, ANSYS Fluent and Mechanical, Root Cause Analysis, PFMEA, Machine Shop Competency, GD&T, Tolerance Stackups

Software: MATLAB, Python, Arduino, Microsoft Office, Google Drive

Languages: Fluent in Spanish; Conversational in Mandarin

INTERNSHIP EXPERIENCE

Symbiotic | Mechanical Engineering Intern | May 2026-August 2026

- Developed a compliant silicone “grass” to prevent packages from rolling on a mobile robot. Created a stable CAD design that could support variations of 300+ features, 3D printed mold design and development, and testing plan.
- Designed a die cast part to replace machined component, resulting in an estimated cost reduction of ~\$15 per part (80%)
- Designed multiple iterations of adhesive gaskets to protect robot from payload spills to minimize need for RTV
- Traveled to customer site to perform mechanical and PCBA reworks on 60+ bots over 4 days.

Locus Robotics | Product Quality Intern | June 2025-August 2025

- Supported in problem statement development and frequency characterizations across 90+ customer sites using robot error reporting and customer reported symptoms (Salesforce). Collated into 8D report for major product escalation.
- Worked cross-functionally with Product Management, Test Engineering, and Data Science to determine the effectiveness of proactive failure monitoring in the field and in-house refurbishing processes.
- Developed work instructions for internal rework process and contract manufacturers.
- Tested and troubleshooted returned bots, PCBAs, LiDARs, and internal bot computers.
- Determined assembly risks by shadowing manufacturing. Generated a PFMEA report out to manufacturing and quality engineers.

WORK EXPERIENCE

Social Decision Science, Carnegie Mellon University | Teaching Assistant | August 2025-Present

- Human Intelligence and Human Stupidity: a lecture-style class discussing theories about intelligence, the relationship between human intelligence and AI, and individual and group intelligence.
- Designing tasks to assess human intelligence and group dynamics for the final exam.
- Creating a custom controller that can be used by five people to play Mario Kart in final assessment.

AiPEX Lab, Carnegie Mellon University | Mechanical Design Research Assistant | February 2025-May 2025

- Designed a self-contained, 5 DOF, tendon driven robotic thumb for a robotic hand and upper arm using SolidWorks.
- 3D printed prototypes and assembled models with off-the-shelf parts.

PROJECT EXPERIENCE

Accessible Keyboard and Mouse Interface | Mechanical Design: Methods and Applications | September 2025-December 2025

- Designed an alternative keyboard and mouse system comprised of a gesture-detecting glove and customizable joystick for users with fine motor impairment under \$200. Enables web navigation through broad hand, arm, or foot movements. [Team of 6]
- Material selection and development of custom bend sensors using piezoresistive fabric
- Construction of wearable glove and interfacing with electronic components

Collapsible Chair | Mechanics 3D | March 2024-May 2024

- Designed, performed FEA, and manufactured a collapsible camping chair using human centered design.
- Chair had a weight of 2.7lb, volume of 3.2L, < 30 second set up time, and simulated FOS of 4.8 for 200lb person.
- Awarded most marketable design out of 36 groups. [Team of 4]